

Methane Index Explorer (MIE) – Estimating Response to Selection for Methane Emissions

Repository URL: <https://github.com/AgResearch/MIE>

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MIE Manual

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Version of R

R version 4.3.0 (2023-04-21 ucrt)

Input Files and Formats

Input File 1: Estimated Genomic Breeding Values (GEBVs)

- **Format:** Excel file (.xlsx)
- **Description:** Contains genomic breeding values for all traits of interest for each animal in the flock. Additionally, there might be columns such as for basic information about the flock (flock number, birth years, breed, etc.). The file should consist of a single table and must include a header row specifying animal IDs and corresponding trait GEBVs in the following format. There is no limit on the number of traits.
- **Example:**

AnimalID,	Trait1_GEBV	Trait2_GEBV	Trait3_GEBV
1001	0.45	1.23	-0.67
1002	0.38	1.11	-0.54

Input File 2: Economic Values of Traits

- **Format:** Excel file (.xlsx)
- **Description:** Contains the economic values assigned to each trait of interest. This file enables users to customize selection indices based on economic goals.
- **Example:**

Trait	Economic Value
Trait1	2.5
Trait2	1.2
Trait3	-0.8

Both files must be properly formatted and checked for consistency before running the analysis. Please note that the first file expects Trait1_GEBV etc for the name and the second file Trait1 only. Also, the trait names must be unique.

Data Sources

Estimated Genomic Breeding Values (GEBVs):

Extracted from the JADE interface of the Sheep Improvement Limited (SIL) database system, which supports genomic and pedigree-based genetic evaluations for livestock.

- Usage: Data modifications

The TXT-formatted data file extracted from the JADE interface must be converted into an Excel file (.xlsx). Additionally, the dagger symbols appearing in some

columns, representing the use of genotype information for estimating GEBVs, must be removed.

- Secondary Data Source: nProve tool

This will provide free accessibility to GEBVs for farmers and breeders.

Economic Values:

These should be based on industry standards, economic analyses, or breeding program-specific goals, tailored to optimize economic returns in livestock breeding. These sources ensure the accuracy and applicability of the analyses provided by the software.

The economic values used in this analysis are based on SIL's published estimates, as provided in the SIL_technical_notes.pdf file.

Output Files and Their Summary

Log File:

- **Format:** TXT
- **Description:** Records basic information about the flock (flock number, total number of animals analyzed from the herd, range of birth years, etc.), traits used for the index, trait economic values, economic values for methane, results from test for normality of methane GEBVs of the flock, and variability estimates of GEBVs.

Selection Response Graph:

- **Format:** PNG
- **Description:** Illustrates the predicted percentage progress of methane and other traits in the index under each economic weight.

Selection Response Table:

- **Format:** CSV
- **Description:** Illustrates the predicted percentage progress of methane and other traits in the index under each economic weight.

Annual Methane Reduction Graph:

- **Format:** PNG
- **Description:** Illustrates the yearly reduction of methane emissions for each economic weight scenario. This assumes generation lengths (L) of 3.5 years for females and 1.3 years for males and selection intensities (i) were estimated considering a selection of 2% of males ($i = 2.421$) and 30% of females ($i = 1.159$), therefore, i/L of 0.75. Users should customize values for i and L based on their herd circumstances.

Density Plot of GEBVs:

- **Format:** PNG
- **Description:** Visualizes the distribution of genomic breeding values of methane.

Histogram of GEBVs:

- **Format:** PNG
- **Description:** Shows the frequency distribution of genomic breeding values of methane across animals.

QQ Plot of GEBVs:

- **Format:** PNG
- **Description:** Compares the distribution of observed GEBVs of methane to the expected distribution to assess normality.